

MUSEUM DOCENT GUIDE · EXHIBIT DEMONSTRATION

The Machine That Flew the Cannonball

How a 1960s analog computer solves a projectile's flight with nothing but voltages, op-amps, and patch cords — and why that problem launched the entire computer age.



Ages 10 → adult

~5–7 min demo

Patch list + troubleshooting

Print-ready

Exhibit: Comdyna GP-6 Analog Computer with Hewlett-Packard 1200B oscilloscope

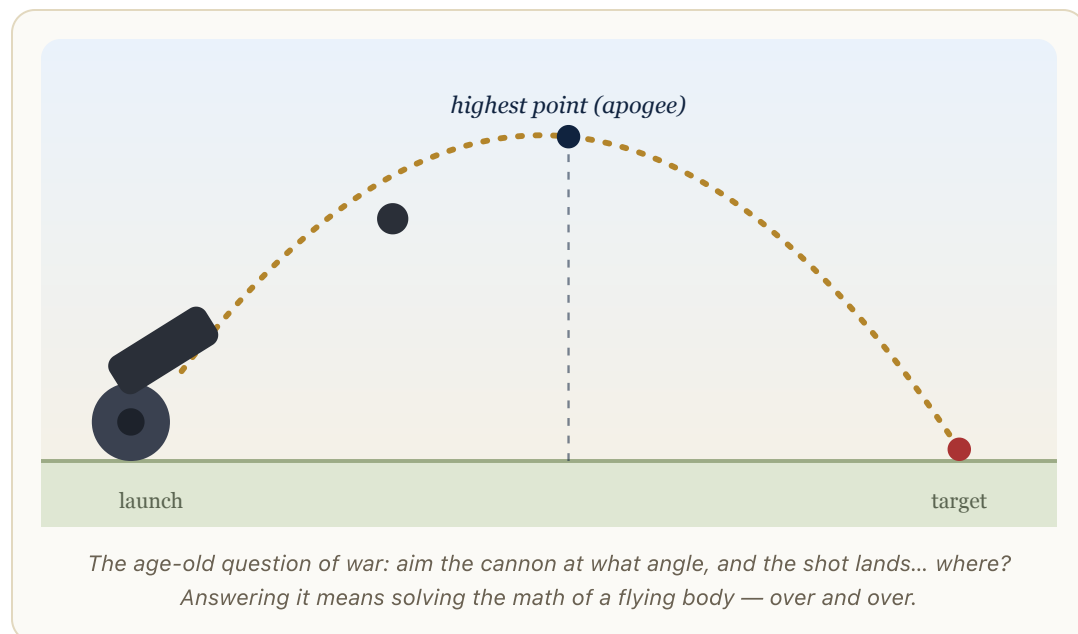
The equation: $d^2y/dt^2 = -g$

SECTION 1

The Big Idea — and why it built the modern world

In one sentence: **this machine has no microchip and runs no software, yet it solves a real physics problem in real time — by turning the problem into electricity.** The green arc on the oscilloscope is a projectile's flight, computed live. Turn a knob and the flight changes shape instantly.

That is strange and wonderful, but it's also *historically enormous*. The problem this exhibit demonstrates — **predicting where a projectile will land** — is, quite literally, the problem that gave birth to the computer industry.



From firing tables to your phone

To hit a target miles away, gunners needed **firing tables** — books listing exactly how to aim for every distance, wind, and powder charge. Each number is the solution to a projectile equation. In World War II the U.S. Army needed these

tables by the thousands, and computing them by hand took teams of people (called “computers”) months per table.

That bottleneck drove the invention of the first electronic computers. **ENIAC** (1945) — often called the first general-purpose electronic computer — was commissioned for exactly this: **calculating artillery firing tables**. Analog computers like the Comdyna GP-6 in front of you attacked the same equations from a different direction — with amplifiers and capacitors instead of code. Both lineages chased the same prize: **knowing where the projectile will go**. Everything downstream — spaceflight, weather models, video games, the device in your pocket — grew from that root.

The throughline for visitors: *“The reason computers exist at all starts with this exact question — and here’s a machine that answers it by becoming the physics, right in front of you.”*

SECTION 2

Explain It at Any Age

Same truth, four altitudes. Pick the one that fits the visitor in front of you — or climb the ladder as they lean in.

Age 10

The curious kid

"See that glowing line? That's a **ball flying through the air**, like throwing a baseball. This old machine is drawing it as it figures out the path. But here's the cool part — there's **no computer chip inside, and no app**. It thinks with **electricity and wires**. Gravity is just a little zap of electricity, and the machine adds it up to figure out how fast the ball goes and how high it gets. Watch — when I turn this knob, I'm changing how strong gravity is, and the whole throw changes! Less gravity, the ball floats way up like on the Moon!"

Age 15

The student

"This is an **analog computer**. Your phone is digital — it solves problems by counting in 1s and 0s, super fast. This one doesn't count at all. It lets a **voltage stand in for a real quantity**: one voltage *is* gravity, one *is* the ball's speed, one *is* its height. You learned in science that if you add up acceleration over time you get speed, and add up speed you get distance. This machine does that adding-up — it's called **integration** — with an electronic part called an integrator. Two integrations turn gravity into the flight path you see."

Age 20

The college mind

"Projectile motion is the differential equation $d^2y/dt^2 = -g$. Rather than solving it numerically in tiny time-steps like a digital computer, this machine solves it **continuously**. Each op-amp-with-a-capacitor is an analog integrator; chain two of them and a constant input (gravity) becomes velocity, then position. The patch cords *are* the program — re-wire them and you've written a new equation. And because it runs in repetitive mode, it re-solves the whole trajectory dozens of times a second, so adjusting a coefficient pot updates the solution in real time. It's a physical, continuous-time computer."

Age 62

The lifelong learner

"You may remember when 'computer' could mean either a digital machine or one of these. Before microprocessors were cheap, engineers designing aircraft, spacecraft, and control systems leaned on analog computers because they'd solve a differential equation **instantly** and let you *feel* the system respond as you turned a dial. This GP-6 is a teaching version of those room-sized machines. The same math that puts this arc on the screen sent Apollo to the Moon and traces back to the WWII firing-table problem that built ENIAC. It trades a little precision for something digital never quite recaptured — intuition you can hold in your hand."

ONE LINE THAT LANDS FOR EVERYONE

"A digital computer **calculates** the motion. This one **becomes** the motion."

SECTION 3 • THE TECHNICAL HEART

From Schematic → Equation → Wiring

This is the path an engineer walks to make an analog computer solve anything: **(1)** write the physics as an equation, **(2)** redraw the equation as a block diagram, **(3)** translate each block into patch cords. When you finish this section, the tangle of wires reads like a sentence.

Step 1 — The physics becomes an equation

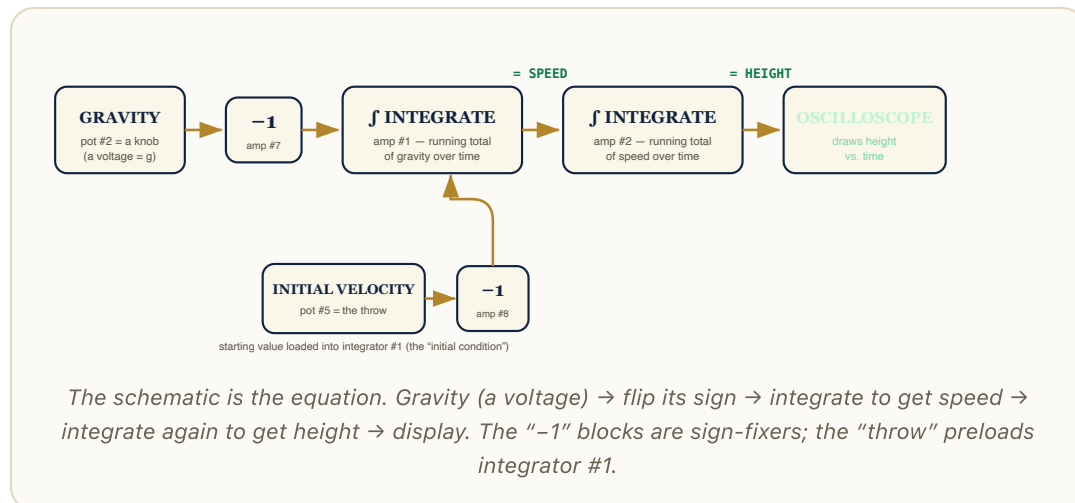
Throw a ball straight up; ignore air. Only gravity acts, a constant downward acceleration **g**. The whole flight is one second-order differential equation:

$$d^2y/dt^2 = -g \quad \text{with start values } y(0)=\text{height}, y'(0)=\text{launch speed}$$

Read it as a chain you can build: **gravity → (add up over time) → speed → (add up over time) → height**. Two “add-ups.” In calculus, adding up over time is **integration**.

Step 2 — The equation becomes a block diagram (the schematic)

Each piece of the equation becomes a labeled block. Follow the signal left to right; the initial “throw” enters the first integrator from below.

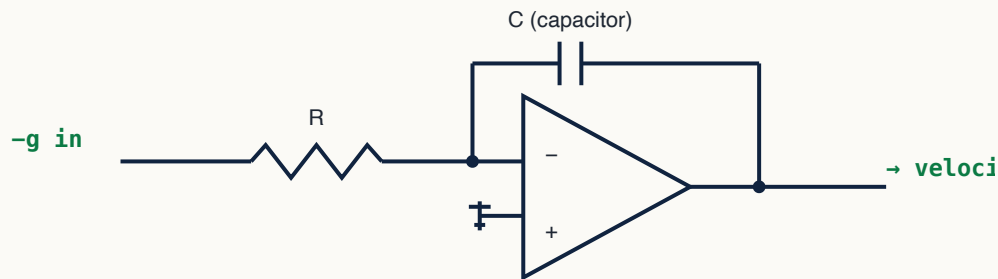


Block	What it is	What it means in $d^2y/dt^2 = -g$
GRAVITY (pot #2)	a knob = a voltage divider	the constant g

-1 (amp #7)	inverter	makes it -g (gravity pulls <i>down</i>)
$\int$ (amp #1)	integrator	first integral → velocity , $v = v_0 - g \cdot t$
INITIAL VELOCITY (pot #5 → amp #8 → amp #1)	a knob + inverter into the integrator's start input	the launch speed v_0
$\int$ (amp #2)	integrator	second integral → position (height)
Oscilloscope	X = time, Y = height	the flight arc — a parabola

What an integrator actually is

The whole machine hinges on one part: an operational amplifier with a **capacitor** in its feedback loop. Its output is the **running total of its input over time** — calculus, done by electronics.



$$\text{output} = -(1/RC) \times \int \text{input } dt + \text{starting value}$$

The integrator. Notice the minus sign in the formula — every integrator flips the sign as it works. That's the only reason the "-1" inverter blocks exist: to keep the physics signs honest.

The key insight that de-tangles the wires: an op-amp integrator computes the *negative* integral. Chain two and the sign flips twice. So the "extra" amplifiers on the schematic (the inverters) aren't doing physics — they're **bookkeeping the minus signs** so velocity falls and the arc points the right way. See that, and the patch panel stops looking like spaghetti.

Step 3 — The block diagram becomes patch cords

On the GP-6 every block's connections are brought out to color-coded jacks. "Programming" means plugging cords so each arrow above becomes a wire. (Exact jack list is in Section 6.) Walking the signal:

1. **+10 V reference → GRAVITY pot (#2)**. Dialing it sets the number **g**.
2. **Pot #2 → inverter (#7)**. Now a steady **-g** — a downward pull.
3. **-g → integrator #1**. Its output ramps; that ramp **is velocity**.
4. **-10 V → INITIAL VELOCITY pot (#5) → inverter (#8) → integrator #1's start input**. Preloads the launch speed.
5. **Integrator #1 → integrator #2**. Output is **height**.
6. **Integrator #2 → oscilloscope Y**. Scope X is time → it draws **height vs. time, the parabola**.

Say it out loud: "Every term in the equation is a block, every arrow is a wire. The schematic is the equation, and the equation is the flight."

SECTION 4

Why the Curve Is Alive — and where “apogee” hides

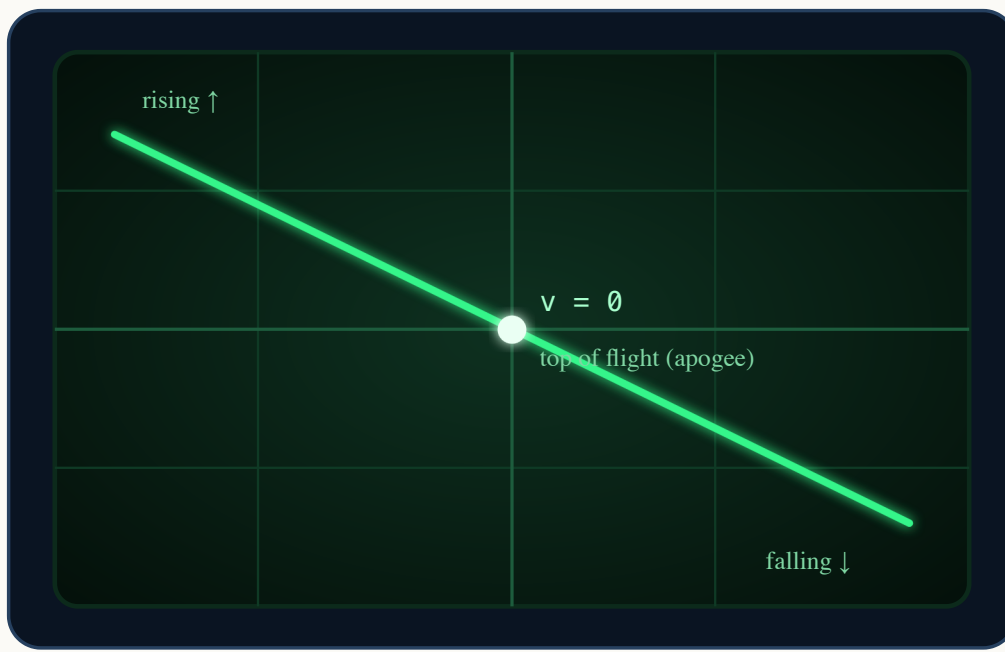
A digital computer steps through tiny slices of time. This machine never steps — its capacitors integrate **continuously**, at the speed of electricity. The whole flight is solved all at once, all the time.

A switch called **Repetitive Operation** makes the machine re-launch the projectile dozens of times a second, redrawing the arc each time. That’s why turning the **GRAVITY** or **INITIAL VELOCITY** knob reshapes the trajectory *instantly and smoothly* — nothing is recompiled; you simply changed a number in a live equation.

Mode	What happens inside	The projectile story
Pot Set	You dial the coefficient knobs and read them on the meter.	Choose gravity and launch speed.
IC (Reset)	Integrators forced to their start values.	The ball is in your hand.
Operate	Integrators integrate; the solution runs.	Launch!
Hold	Integration freezes.	Pause the ball mid-air.
Repetitive Op	Auto-cycles Reset → Operate many times/second.	Re-flies it forever → a stable, live trace.

The hidden gem: find the top of the flight

Switch the scope to view **velocity** instead of height and the parabola becomes a **straight line sloping down**. Gravity steadily steals speed. Where that line **crosses zero**, the ball has stopped rising — that instant is the **apogee**, the very top. After it, the ball falls.



Velocity vs. time. The straight, falling line is gravity bleeding off speed. The zero-crossing is the apogee — a beautiful “aha” for any age.

SECTION 5 • FOR THE PRESENTER

The Demo Script

A timed walkthrough — about 5–7 minutes. *Stage directions are in green italics.*
Spoken lines are in quotes. Land the **bold** beats.

0 • The 60-second hook for walk-ups

[Point at the parabola.] "See that curve? That's a ball flying through the air — being figured out **right now** by this machine. And here's the surprise: **no chip, no software**. It's an **analog** computer — it uses **electricity to act out the physics**. Gravity is a voltage set by this knob. Watch." *[Turn GRAVITY slowly.]* "**The whole trajectory reshapes — instantly**. Less gravity, it sails like the Moon; more, it slams down. I didn't re-run anything. The wires just *are* the new equation. **That's the charm of analog computing.**"

1 • The hook 0:30

[Beside the scope, parabola visible.] "Look at this green arc — a ball thrown across a field, up, over, and down. **This machine is drawing it because it's computing the flight in real time.** And it's a computer with **no chip, no code, no software.** It's called an **analog computer.**"

2 • Digital vs. analog 1:00

"Your phone is **digital** — it solves problems by **counting**, billions of 1s and 0s a second. This one is **analog**. It doesn't count — it uses **voltage to stand in for real things**. A voltage here *is* gravity, one *is* the ball's speed, one *is* its height. It's **wired** so the electricity obeys the same physics a real ball obeys. **It doesn't calculate the motion — it becomes the motion.**"

[Gesture at the patch cords.] "Those wires are the program."

3 • The physics, in three steps 1:15

[Mime a ball arcing.] "One thing acts on a thrown ball: **gravity**, pulling down steadily. So — **one**, gravity is a steady voltage, set by this **GRAVITY** knob. *[Tap pot #2.]* **Two**, add that pull up over time and you get **speed** — the block that adds up is an **integrator**. **Three**, add the *speed* up over time and you get **height** — that's the arc. Gravity → integrate → speed → integrate → height."

4 • Read the wiring like a sentence 1:00

[Trace the schematic / panel with a finger.] "Start at the **GRAVITY** knob → into the **first integrator** → out comes **speed**. This other knob, **INITIAL VELOCITY**, is how hard we throw it. *[Tap pot #5.]* Into the **second integrator** → out comes **height** → to the **scope**. That's your arc. Those little 'invert' amps don't do physics — they just keep plus and minus straight so gravity pulls down and the throw goes up."

5 • The payoff — make it live 1:30

"Here's the magic — it re-flies the ball dozens of times a second, so I can change the world **mid-flight**." *[Turn GRAVITY down slowly.]* "Weaken gravity... the arc **stretches** — that's the **Moon**." *[Turn it up.]* "Crank it up — the arc **collapses**. Heavy planet." *[Turn INITIAL VELOCITY up.]* "Throw harder — bigger arc. **I never typed anything. The trajectory changes the instant I turn the knob, because the wiring is the equation.**" *[Optional: switch to the velocity view.]* "This falling line is **speed**; where it crosses zero is the **top of the flight — the apogee**."

6 • The close 0:30

"For about thirty years — through the Apollo program, aircraft and missile design — **this** is how engineers solved the hardest equations: not by counting, but by building a little electrical model and watching it run. Digital won on precision and flexibility. But for *feeling* how a system behaves, nothing beats it. **That's the charm of analog computing.**"

PRESENTER TIPS

- Turn knobs **slowly** — the magic is watching the curve *morph*.
- Don't re-patch cords mid-tour unless confident (Section 6 restores it).
- If the trace is flat/off-screen, you're likely in Reset/Hold or a pot is at zero — confirm Operate/Repetitive and raise the pots.

The one line to never skip: *"It doesn't calculate the motion — it becomes the motion."*

Operator Setup & Patch List

Authority: the museum's tan schematic handout is the master wiring diagram for this exact patch. If anything here disagrees with the physical panel or that sheet, **trust the sheet**. The list below reconstructs it block-for-block — verify before re-patching.

A • Power-up & display

1. Power on the GP-6 and the HP 1200B; warm up ~1 minute.
2. Scope: computed signal on **vertical (Y)**, **time on horizontal (X)**; start ~1 **V/div**, set the sweep so one full flight fills the screen; adjust intensity/focus.
3. Confirm the red **OVLD** lamp is **off**. If lit, an amplifier is railed — lower the GRAVITY / INITIAL-VELOCITY pots and recheck patching.

B • Front-panel settings (the photographed configuration)

Control	Setting	Why
Mode selector	OPR (operate)	Lets the run modes drive the solution.
Run mode	RO (repetitive)	Auto-cycles Reset→Operate → live, stable trace.
Compute Time	mid (~ 20–50)	Repeat rate; tune so the arc is steady and fills the screen.
GRAVITY — pot #2	~ 0.16 (≈ -0.160 on meter)	The value of g. The “hero” knob.
INITIAL VELOCITY — pot #5	~ 0.66 (≈ -0.660 on meter)	Launch speed. The second demo knob.

Meter reads **percent of 10 V** ($-0.588 = -5.88 \text{ V} = -59\%$). Panel address tape: **X = TIME**; **Y/Pot 7 = GRAVITY**, **5 = G×V**; **Solution 3 = VELOCITY**, **2 = POSITION**.

C • Patch list (block-for-block)

Each amplifier brings out **SJ** (input / summing junction), **IC** (start-value input), and **output**. Pots have input + wiper. References are $\pm 10\text{ V}$.

#	From	To	Represents
1	+10 V ref	Pot #2 in (GRAVITY)	scale into g
2	Pot #2 out	Amp #7 in (invert)	make $-g$
3	Amp #7 out	Amp #1 SJ (integrate)	$-g \rightarrow$ velocity
4	-10 V ref	Pot #5 in (INIT VEL)	scale into v_0
5	Pot #5 out	Amp #8 in (invert)	make $+v_0$
6	Amp #8 out	Amp #1 IC	preload launch speed
7	Amp #1 out	Amp #2 SJ (integrate)	velocity $\rightarrow$ height
8	Amp #2 out	Scope Y	height vs. time (parabola)
9	Amp #1 out	Amp #3 in (invert)	flip velocity for display
10	Amp #3 out	Scope Y (alt.)	velocity line
—	Time base	Scope X	time axis

Grounds: black cords tie scope and computer references together — don't pull them thinking they're spare.

D • Run sequence (each demo)

1. Mode $\rightarrow$ **Pot Set**; confirm GRAVITY (#2) and INITIAL VELOCITY (#5) on the meter; back to **OPR**.
2. Ensure **RO**; a steady arc should appear.
3. Demo **GRAVITY** (slow): arc stretches (Moon) / collapses (heavy planet).
4. Demo **INITIAL VELOCITY**: bigger / smaller arc.
5. Freeze for discussion with **Hold**; resume with **Operate/RO**.

E • Troubleshooting

Symptom	Likely cause	Fix
OVLD lit; trace pinned to edge	pot too high / mis-patch	Lower pots; verify cords 1–7 vs. the sheet.
Flat line, no arc	in Reset/Hold, pot at zero, or scope on wrong node	Set Operate/RO; raise pots; scope Y on Amp #2.

Arc drifts / won't sit still	Compute Time or sweep mismatched	Re-tune Compute Time & scope time/div.
Curve frozen, ignores knobs	stuck in Hold, or in Pot Set	Return to Operate + RO.
Wrong trace (line vs. parabola)	scope Y on wrong amp	Amp #2 = parabola; Amp #3 = velocity line.

From Cannonballs to Your Phone

- **1600s** — Galileo shows a projectile's path is a **parabola**; the math of flight begins.
- **1700s–1800s** — Armies compile **firing tables** by hand; "computers" are people doing arithmetic.
- **1930s–40s** — Analog **differential analyzers** mechanize the equations of motion for ballistics and engineering.
- **1945** — **ENIAC** is built to compute artillery firing tables — a founding moment of digital computing.
- **1950s–70s** — Analog computers like this GP-6 design aircraft, spacecraft, and control systems; Apollo-era engineers live on them.
- **Today** — The same equations run on the chip in your pocket. It all traces back to one question: **where will the projectile land?**

Comdyna GP-6 Projectile Demonstration — Docent Guide. Prepared for the museum exhibit pairing the GP-6 analog computer with the HP 1200B oscilloscope. The patch list reconstructs the museum's tan schematic block-for-block; verify against the physical panel and that sheet before re-patching. · Print on US Letter; colors and diagrams are tuned for color printing.